CONVOSHRIMP

ConvoShrimp: Biological vs. Artificial Intelligence

The Mantis Shrimp as a Model for Convolutional Neural Networks

A02 Neural Network Zoo

ITAI 2376 Deep Learning Artificial Intel

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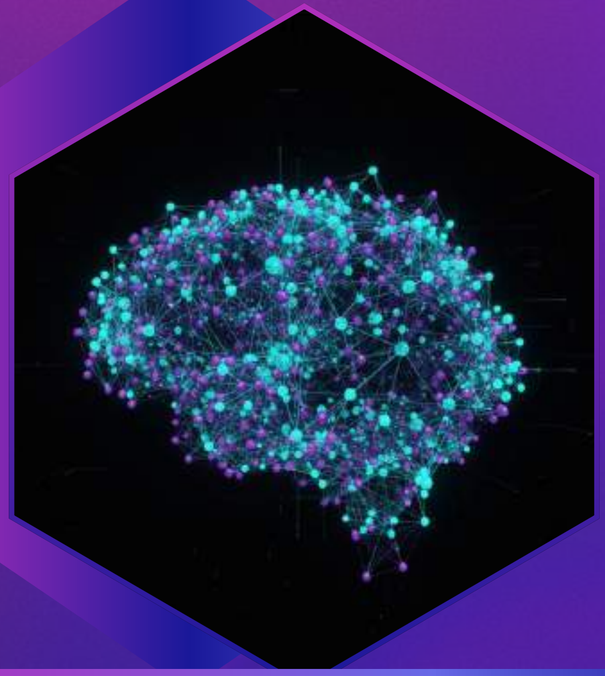
INTRODUCTION

The Biological Supercomputer

- **Biology:** The Mantis Shrimp (Odontodactylus scyllarus)
- **Technology:** Convolutional Neural Networks (CNNs)
- **The Connection:** Nature's solution to high-speed computer vision

Description: The Peacock Mantis Shrimp possesses one of the most complex visual systems in the animal kingdom, processing massive visual data with near-zero latency. Convolutional Neural Networks function as the "visual cortex" of modern AI, engineered to recognize patterns and process images efficiently.



WHAT IS A NEURAL NETWORK?

Key Concepts:

- **Neuron:** Basic unit of computation
- **Layers:** Input → Hidden → Output
- **Learning:** Automated weight adjustment through error minimization

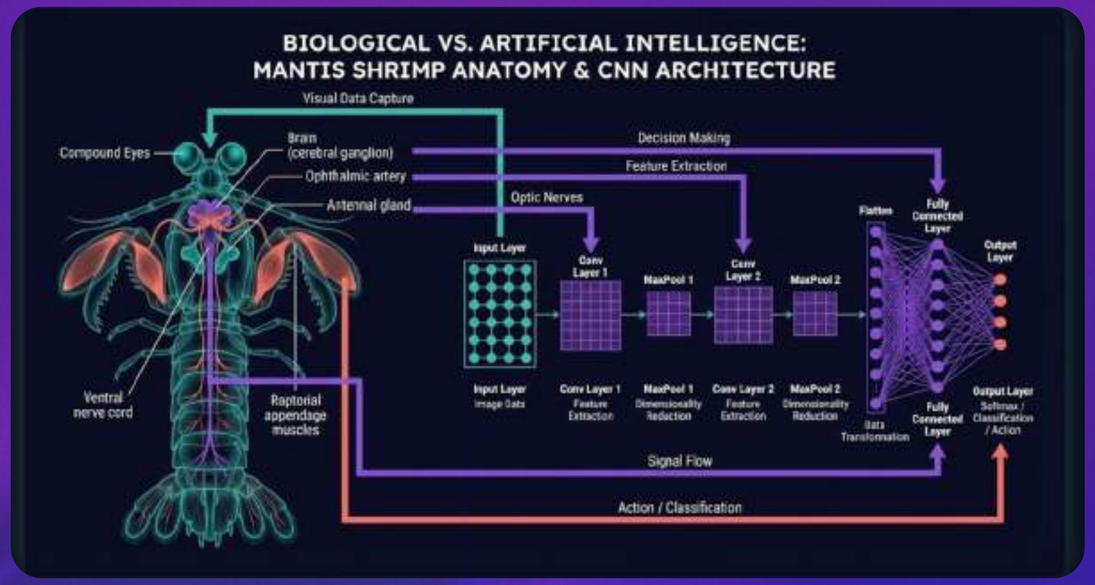
Description: A neural network is a computational model inspired by the biological brain. Interconnected layers of artificial neurons transmit and transform data. Unlike rule-based algorithms, neural networks learn pattern recognition by processing datasets and adjusting internal parameters to minimize prediction errors.

ODONTODACTYLUS SCYLLARUS

Specifications:

- **Habitat:** Indo-Pacific coral reefs
- **Vision:** 16 photoreceptor channels (humans have 3)
- **Speed:** Strike velocity of 23 m/s
- **Processing:** Edge computing (local processing in the eye)

Description: The Peacock Mantis Shrimp is a predatory crustacean with a decentralized visual processing architecture. Visual information undergoes partial local processing within the eye stalk before brain transmission a biological analogue to edge computing in efficient AI systems.



VISION COMPARISON — 3 CHANNELS VS. 16 CHANNELS

Human Vision (RGB):

- **3 Channels:** Red, Green, Blue
- **Spectrum:** 400nm – 700nm

Mantis Shrimp Vision:

- **16 Channels:** 12 Color + 4 Polarization
- **Spectrum:** 300nm (Deep UV) – 720nm (Far Red)
- **Trinocular Depth:** Independent depth perception per eye

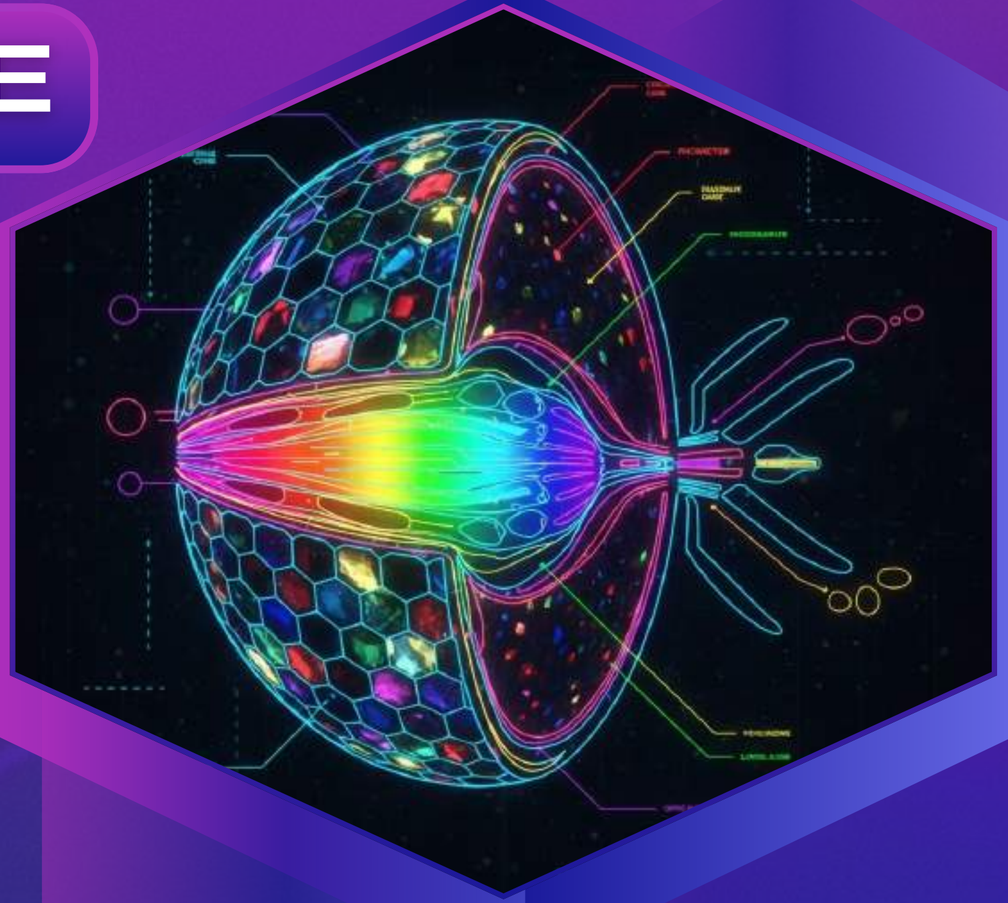
Description: Human vision and standard cameras rely on 3 color channels (RGB). The Mantis Shrimp utilizes 16 distinct channels including UV, IR, and polarized light receptors. In AI terms, this equals hyperspectral imaging detecting features invisible to standard RGB sensors, such as underwater camouflage or biological anomalies.

THE COMPOUND EYE

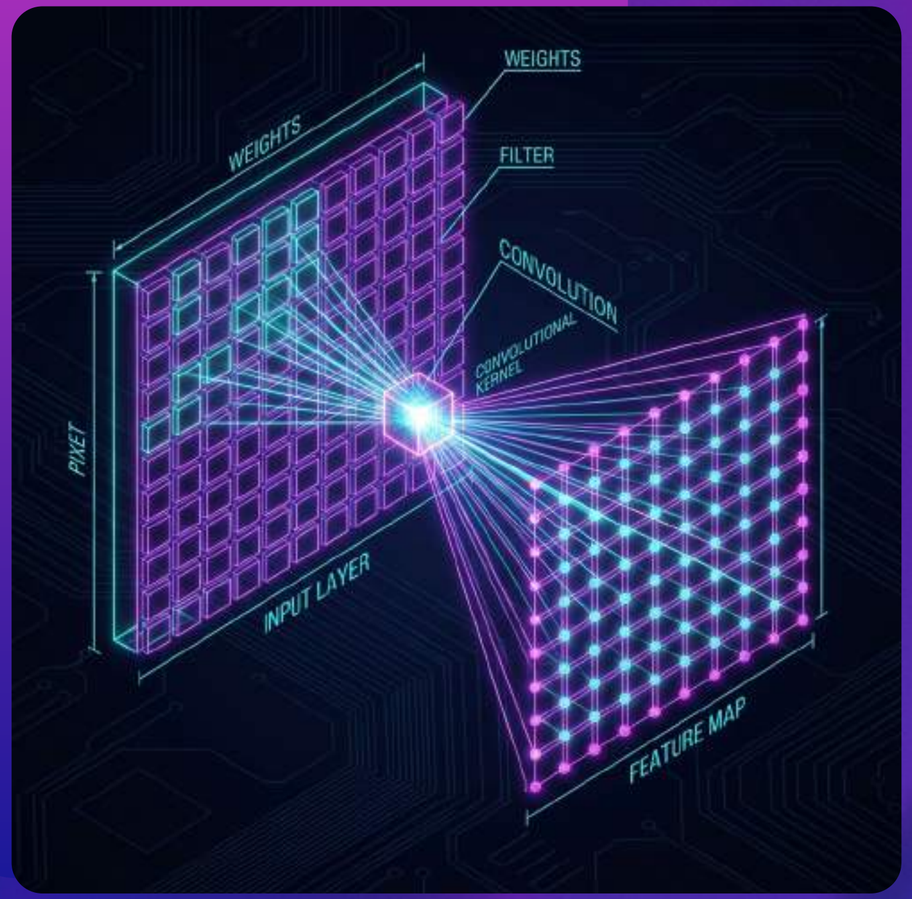
Structural Components:

- **Ommatidia:** Thousands of individual hexagonal lenses
- **The Midband:** Specialized central strip for color and polarization analysis
- **Hexagonal Grid:** Optimized sensor packing density

Description: The mantis shrimp eye is a compound structure of thousands of hexagonal ommatidia units. The Midband a specialized central strip with six rows of enhanced photoreceptors functions as a dedicated hardware accelerator for color and polarization analysis, separate from motion-detecting hemispheres.



HOW CNNS "SEE"



Processing Pipeline:

1. **Input:** Raw pixel data matrix
2. **Convolution:** Feature extraction via filter scanning
3. **Pooling:** Dimensionality reduction
4. **Output:** Classification probability

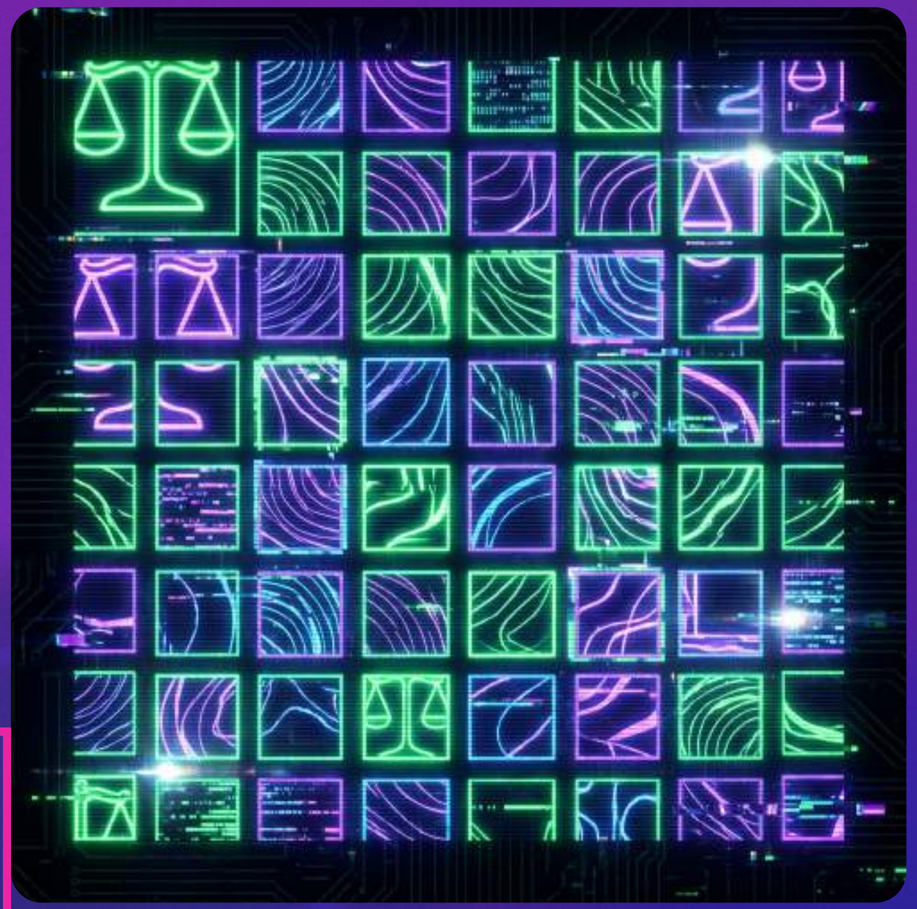
Description: Convolutional Neural Networks mimic biological vision through layered processing. The network scans input images with small filters (kernels) to detect low-level features like edges and textures. Subsequent layers combine these into complex shapes and objects, reconstructing semantic understanding from raw pixel data.

SPEED & PRECISION

Performance Factors:

- **Biological Latency:** Millisecond reaction time
- **The Challenge:** Real-time visual processing requirements
- **AI Solution:** Hardware acceleration (GPUs/TPUs)
- **Application:** Autonomous vehicle safety systems

Description: The mantis shrimp strikes at 23 m/s, requiring visual processing in milliseconds any latency means a missed target. This biological constraint parallels autonomous vehicle requirements: instant obstacle detection and braking execution. Modern AI uses GPU acceleration for necessary low-latency throughput.



REAL-WORLD APPLICATIONS

Application Areas:

- **Medical Imaging:** Hyperspectral tumor detection via UV/IR signatures
- **Satellite Monitoring:** Agricultural health tracking
- **Underwater Robotics:** Polarization-based navigation in turbid water

Description: Bio-inspired multi-channel vision systems operate in critical fields. Medical imaging uses hyperspectral analysis to differentiate tumors from healthy tissue. Underwater robotics utilize polarization sensors mimicking mantis shrimp capabilities, to navigate where standard visual cameras fail.

NATURE IS THE ULTIMATE ENGINEER

Key Takeaways:

- Millions of years of evolutionary R&D
- Proven blueprints for efficiency, speed, and architecture
- **Future direction:** Bio-inspired AI hardware

Description: Biology represents millions of years of evolutionary research and development. The mantis shrimp's architecture provides a proven blueprint for efficient, high-speed, multi-spectral visual processing. Studying these biological systems enables development of next-generation AI hardware faster, more efficient, and more capable than current silicon-based approaches

WEBSITE LINK

All addtional inforamtion and references in depth in the website below check it out. Interactive lookbook showcasing Convolutional Neural Networks as a Mantis Shrimp.

ConvoShrimp - CNN Neural Network Lookbook
https://convoshrimp.netlify.app/

Video: The Mantis Shrimp as a Model for Convolutional Neural Networks
Note: AI video by Google Gemini Veo 3

Mantis_Shrimp_D_Rotational_Video_Showcase.mp4

Thank You